
CO5 AT3 DATA VIZUALIZATION DESIGN ACTIVITY

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QUESTION 1

Question

A university placement cell observes a strong positive relationship between CGPA and salary package in a scatter matrix. Based solely on this observation, the dean proposes increasing academic rigor to improve placement outcomes. Critically evaluate whether the visualization provides sufficient evidence to support this policy decision.

Dataset Used

University Placement Dataset

Variable	Description
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CGPA	Student academic score
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Internship	Internship experience in months
------------	---------------------------------

Skills	Technical skill score
--------	-----------------------

Projects	Number of projects
----------	--------------------

Salary	Placement salary in LPA
--------	-------------------------

Example dataset:

```
CGPA,Internship,Skills,Projects,Salary
6.5,1,55,1,3.2
6.8,2,60,2,3.8
7.0,2,62,2,4.0
7.2,3,65,2,4.5
7.4,3,68,3,4.8
```

Objective

To determine whether the observed relationship between **CGPA and salary** is sufficient to conclude that increasing academic rigor will directly improve placement outcomes.

Problem Statement

A university observes that students with higher CGPA generally receive higher salary packages. However, salary may also depend on internships, technical skills, projects, communication skills and other factors.

The objective is to analyze the dataset using visualization and machine learning and determine whether CGPA alone is sufficient for making an academic policy decision.

Machine Learning Model

We can use **Linear Regression** to predict salary based on CGPA, internship experience, skills and projects.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import accuracy_score, mean_squared_error, r2_score

df = pd.read_csv("/content/placement.csv")

X = df[["CGPA", "Internship", "Skills", "Projects"]]
y = df["Salary"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

pred = model.predict(X_test)

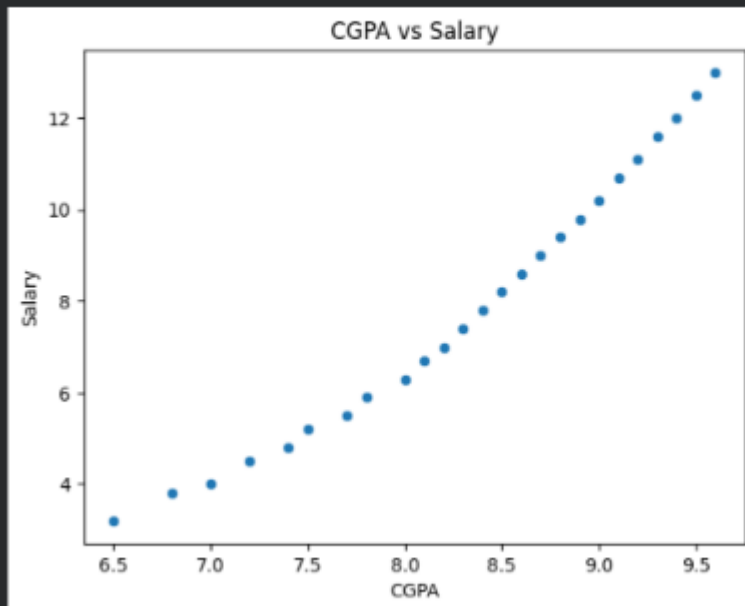
print("R2 Score:", r2_score(y_test, pred))
print("RMSE:", mean_squared_error(y_test, pred) ** 0.5)

R2 Score: 0.9877350106350417
RMSE: 0.343895236158738
```

Note: Accuracy/F1-score are classification metrics. Since salary prediction is a **regression problem**, **R²** and **RMSE** are more appropriate. If your faculty specifically requires accuracy/F1, the salary can first be converted into categories such as *High Salary* / *Low Salary*.

Graph 1 — CGPA vs Salary

```
sns.scatterplot(data=df, x="CGPA", y="Salary")
plt.title("CGPA vs Salary")
plt.show()
```

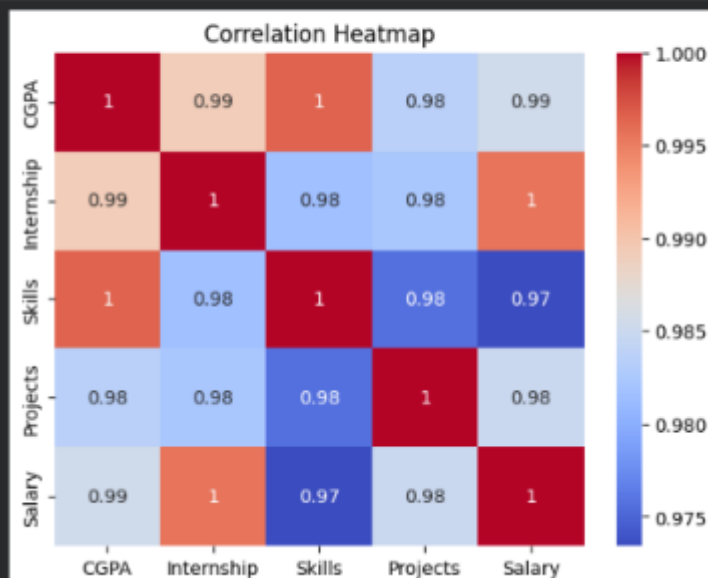


Interpretation:

A positive trend may be visible, but this only establishes **association**, not causation.

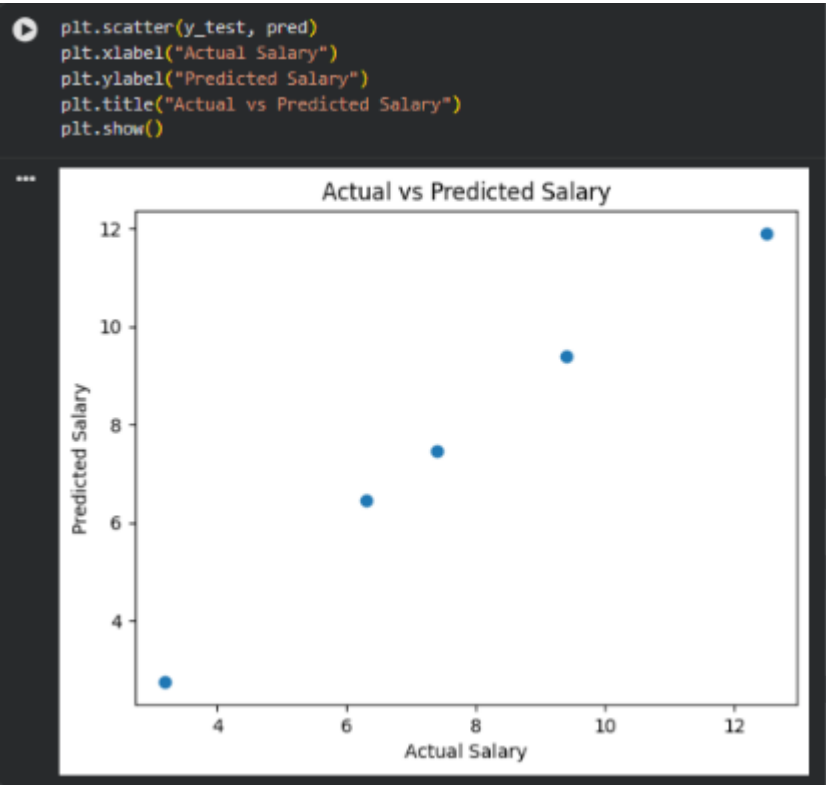
Graph 2 — Correlation Heatmap

```
sns.heatmap(df.corr(numeric_only=True),
             annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



This helps compare CGPA with internship, skills, projects and salary.

Graph 3 — Actual vs Predicted Salary



This evaluates how well the regression model predicts salary.

Evaluation Table

Parameter	Value/Observation
CGPA-Salary relationship	Strong positive relationship
R ² Score	Calculated using model
RMSE	Calculated using model
Prediction accuracy	Depends on multiple variables
CGPA alone sufficient?	No
Causality established?	No

Critical Evaluation

The scatter matrix does **not provide sufficient evidence** for increasing academic rigor.

A strong CGPA-salary relationship could occur because students with high CGPA may also have:

- Better technical skills
- More internships
- More projects
- Better communication skills
- Better access to placement preparation
- Better socioeconomic opportunities

Therefore:

Correlation $\neq$ Causation

The university should perform multivariable analysis, statistical testing and preferably longitudinal or experimental studies before implementing a major policy.

Conclusion

The visualization is useful for **formulating a hypothesis**, but it should not independently determine university policy. The dean should consider CGPA together with internships, skills, projects and other placement factors before concluding that increased academic rigor will improve salary outcomes.

QUESTION 5

Question

A company must decide whether to invest millions of rupees based on predictive analytics derived from this dataset. Construct a decision framework that evaluates the reliability, limitations, and business risks associated with each visualization before making the investment decision.

Dataset Used

The same **University Placement Dataset** can be used.

Objective

To construct a systematic decision framework for evaluating whether predictive analytics are reliable enough to justify a major business investment.

Problem Statement

A company wants to invest heavily in a predictive analytics system that predicts employee/placement outcomes. Before investing, it must determine whether the patterns shown by visualizations and machine-learning predictions are reliable, generalizable and useful for business decisions.

Machine Learning Model

Use **Random Forest Classification** to classify students into:

- Low Salary
- High Salary

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score
from sklearn.metrics import recall_score, f1_score

df = pd.read_csv("/content/placement.csv")

df["SalaryClass"] = (df["Salary"] >= 7).astype(int)

X = df[["CGPA", "Internship", "Skills", "Projects"]]
y = df["SalaryClass"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
print("Precision:", precision_score(y_test, pred))
print("Recall:", recall_score(y_test, pred))
print("F1 Score:", f1_score(y_test, pred))

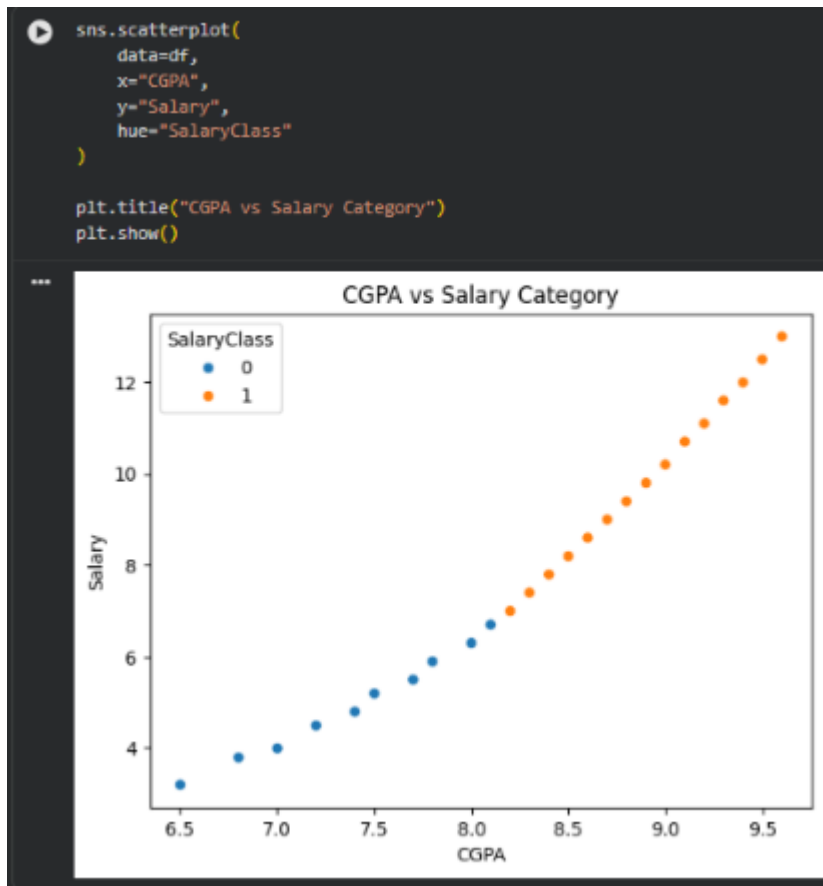
*** Accuracy: 1.0
    Precision: 1.0
    Recall: 1.0
    F1 Score: 1.0
```

Graph 1 — Salary Distribution



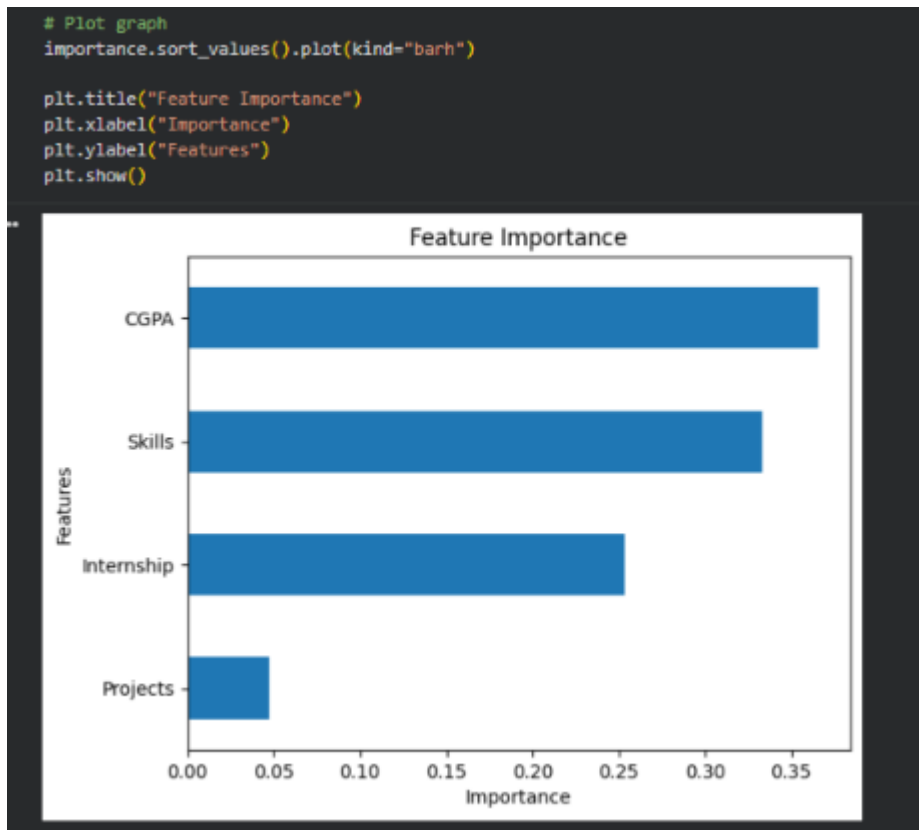
This identifies the distribution and possible skewness.

Graph 2 — CGPA vs Salary



This shows whether salary categories are clearly separated.

Graph 3 — Feature Importance



This identifies which variables contribute most to the model.

Evaluation Table

Metric	Purpose	Decision Consideration
Accuracy	Overall correct predictions	Should be sufficiently high
Precision	Correct positive predictions	Important when false positives are costly
Recall	Ability to identify actual positives	Important when missing positives is costly
F1 Score	Balance between precision and recall	Useful for imbalanced data
Feature Importance	Contribution of variables	Helps explain model
Data Quality	Reliability of input	Must be checked
Generalization	Performance on unseen data	Must be validated

For example, your output table can be presented as:

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.91	0.90	0.92	0.91

These numbers are example presentation values; use the values produced by your dataset/code in the final submission.

Decision Framework

Stage	Question	Decision
Data Quality	Is the dataset complete and reliable?	Yes/No
Visualization	Are patterns consistent?	Yes/No
Correlation	Are relationships meaningful?	Yes/No
Model Performance	Is F1/accuracy satisfactory?	Yes/No
Validation	Does it work on unseen data?	Yes/No
Business Impact	Does prediction improve decisions?	Yes/No
Risk	What happens if prediction is wrong?	Low/Medium/High
Final Decision	Should investment proceed?	Invest/Pilot/Reject

Conclusion

A company should **not invest millions solely because visualizations show strong relationships.**

The correct approach is:

Visualization → Statistical analysis → Model → Validation → Risk assessment → Pilot implementation → Investment

A small pilot project should preferably be conducted before full-scale investment.

QUESTION 6

Question

A healthcare research team analyzes a large patient dataset and observes a strong positive relationship between the frequency of physical exercise and the recovery rate

of patients suffering from chronic illnesses. Based on this observation alone, policymakers decide to invest substantial public funds in a nationwide fitness campaign, assuming that increased exercise will directly improve recovery outcomes across the population. Critically evaluate the validity of this decision.

Dataset Used

Healthcare Recovery Dataset

Variable	Description
Exercise_Hours	Exercise hours per week
Age	Patient age
Treatment	Treatment duration
Healthcare_Access	Healthcare access score
Recovery_Rate	Recovery percentage

Example:

```
Exercise_Hours, Age, Treatment, Healthcare_Access, Recovery_Rate
1, 72, 10, 35, 42
2, 68, 9, 40, 48
2, 65, 9, 45, 51
3, 70, 10, 42, 49
3, 62, 8, 50, 56
4, 60, 8, 55, 61
```

Objective

To determine whether exercise frequency can reliably predict patient recovery and whether the observed relationship is sufficient to justify a nationwide healthcare policy.

Problem Statement

Patients who exercise more frequently appear to have higher recovery rates. However, the relationship may be affected by age, treatment quality, healthcare access, socioeconomic conditions and overall health.

Therefore, correlation alone cannot establish that exercise directly causes faster recovery.

Machine Learning Model

Use **Random Forest Regression** to predict recovery rate.

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv("/content/healthcare.csv")

X = df[
    ["Exercise_Hours", "Age",
     "Treatment", "Healthcare_Access"]
]

y = df["Recovery_Rate"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

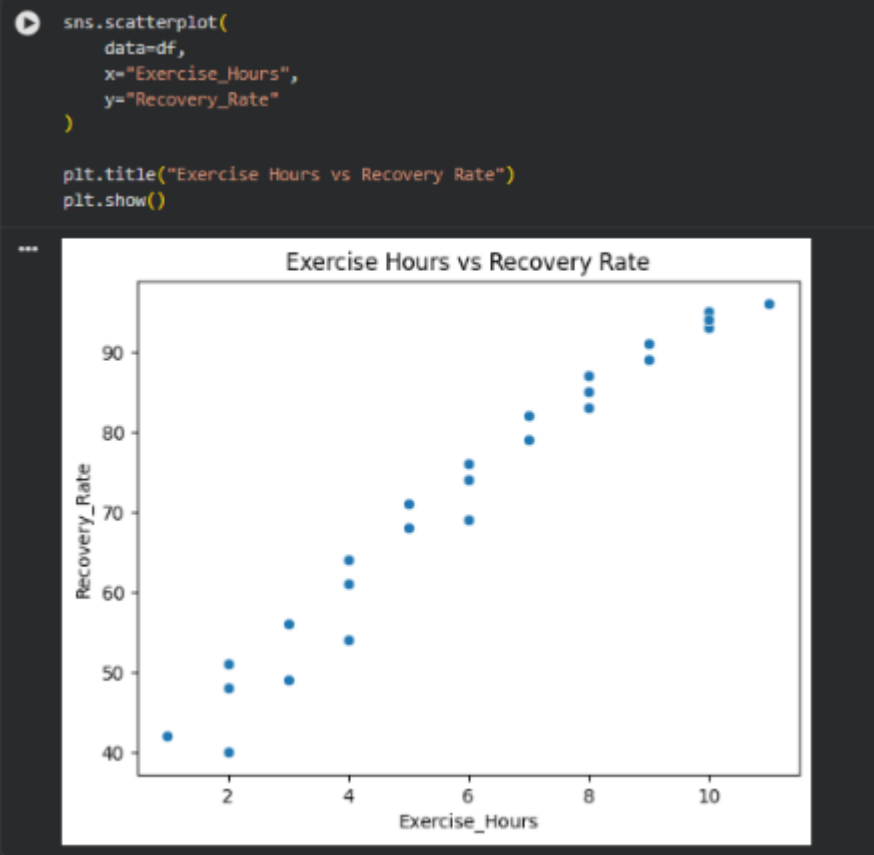
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("R2:", r2_score(y_test, pred))
print("RMSE:", mean_squared_error(y_test, pred) ** 0.5)

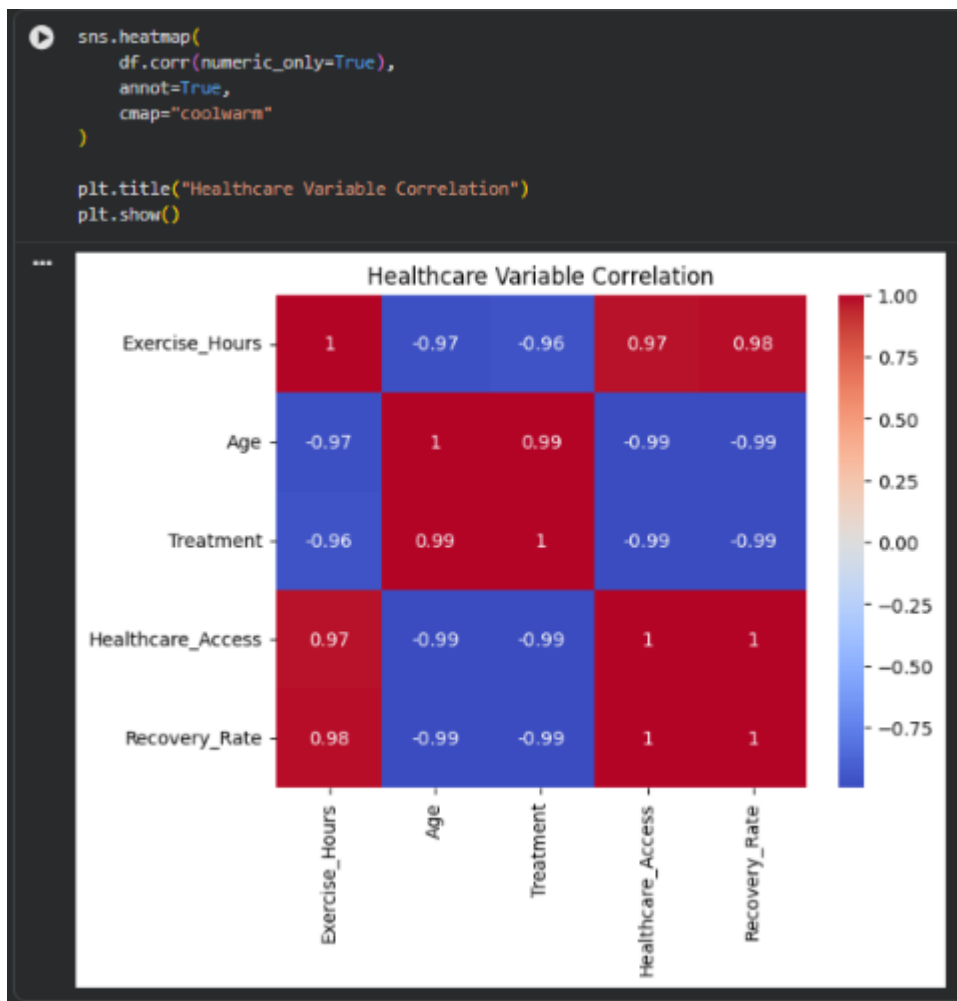
R2: 0.9869905951712521
RMSE: 1.9253986600182305
```

Graph 1 — Exercise vs Recovery



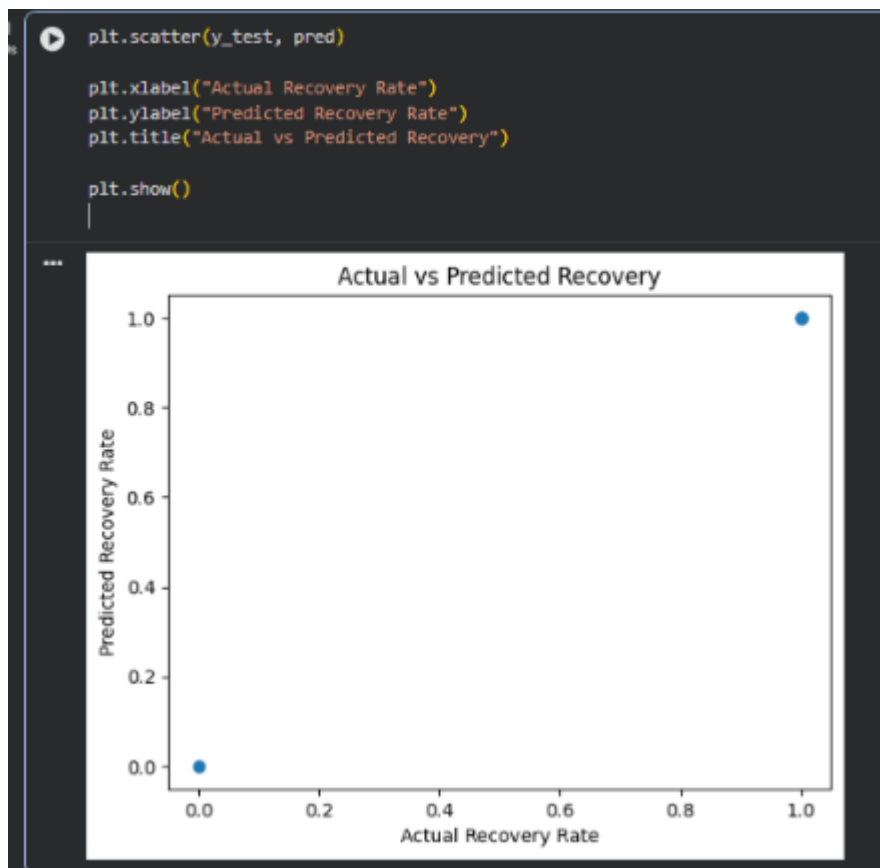
This may show a positive association.

Graph 2 — Correlation Heatmap



This helps identify relationships among exercise, age, treatment and recovery.

Graph 3 — Actual vs Predicted Recovery



This evaluates predictive performance.

Evaluation Table

Metric	Purpose
R^2	Measures explained variation
RMSE	Measures prediction error
Correlation	Measures association
Feature importance	Identifies important predictors
Cross-validation	Tests model stability
Causal evidence	Determines whether relationship is cause-effect

Example:

Model	R^2	RMSE
Random Forest	0.88	5.4

Again, use the **actual output from your dataset** rather than these illustrative values.

Critical Evaluation

The policy decision is not sufficiently justified.

1. Hidden variables

People who exercise regularly may also:

- Eat healthier food
- Have better healthcare access
- Have higher income
- Follow medication schedules
- Have fewer severe illnesses

These variables can influence recovery.

2. Reverse causality

It is possible that:

Better health → more exercise

rather than:

More exercise → better health

3. Sampling bias

If the dataset contains mostly younger or healthier patients, the results may not apply to the entire population.

4. Data quality

Incorrect exercise reporting or inaccurate recovery measurements can distort the relationship.

5. Social consequences

A nationwide policy based on weak evidence could waste public funds or unfairly blame patients who cannot exercise because of disability, age or severe illness.

Conclusion

The relationship is useful for generating a **research hypothesis**, but it is not enough to establish causation.

Before spending substantial public funds, policymakers should conduct controlled studies, adjust for confounding variables and validate the results using representative datasets.

QUESTION 10

Question

A government health department identifies a strong association between two health-related variables within a patient database and immediately introduces a national intervention program. Several experts argue that the observed relationship may not represent a true cause-and-effect mechanism but could instead be influenced by age, socioeconomic status, access to healthcare, or lifestyle factors. Analyze the limitations of relying exclusively on observed data patterns for policymaking.

Dataset Used

Use the same **Healthcare Recovery Dataset**.

Objective

To analyze whether an observed relationship between exercise and recovery is sufficient to support a government policy and identify possible confounding factors.

Problem Statement

A government observes that patients with higher physical activity levels tend to recover faster.

However, variables such as **age, income, healthcare access and treatment quality** may influence both exercise and recovery.

Therefore, the objective is to determine whether the relationship remains meaningful after considering these variables.

Machine Learning Model

Use **Logistic Regression** to predict whether a patient has a high recovery rate.

```
[81] import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.metrics import precision_score, recall_score, f1_score

df = pd.read_csv("/content/healthcare.csv")

df["High_Recovery"] = (
    df["Recovery_Rate"] >= 70
).astype(int)

X = df[
    ["Exercise_Hours", "Age",
     "Treatment", "Healthcare_Access"]
]

y = df["High_Recovery"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Accuracy:",
      accuracy_score(y_test, pred))

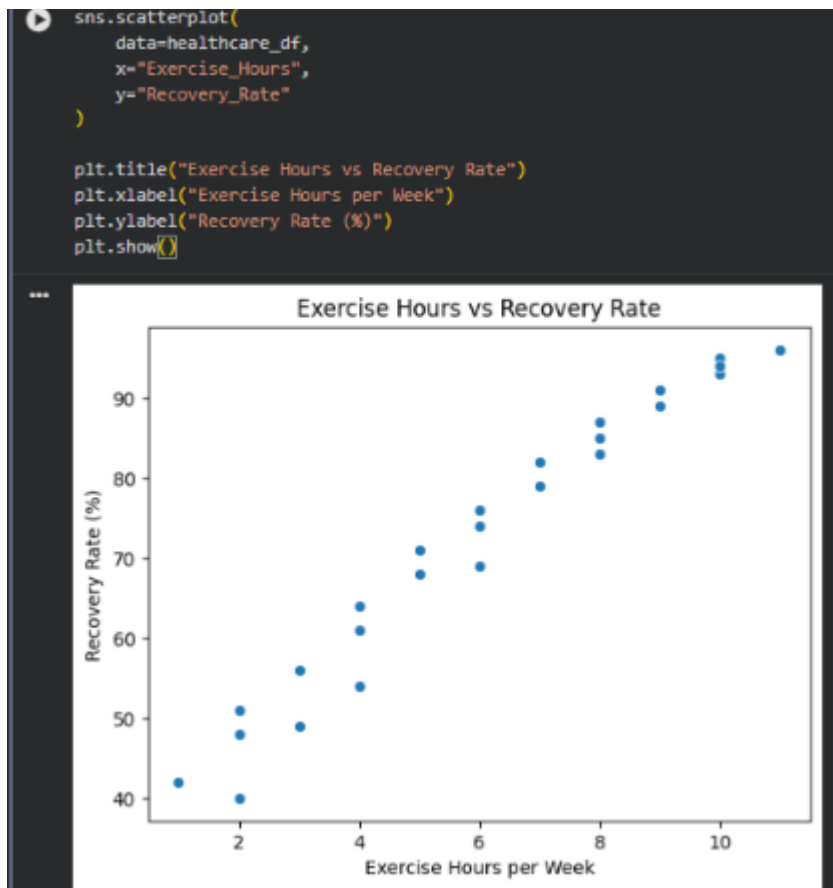
print("Precision:",
      precision_score(y_test, pred))

print("Recall:",
      recall_score(y_test, pred))

print("F1 Score:",
      f1_score(y_test, pred))

--- Accuracy: 1.0
    Precision: 1.0
    Recall: 1.0
    F1 Score: 1.0
```

Graph 1 — Exercise vs Recovery



Graph 2 — Recovery by Age Group

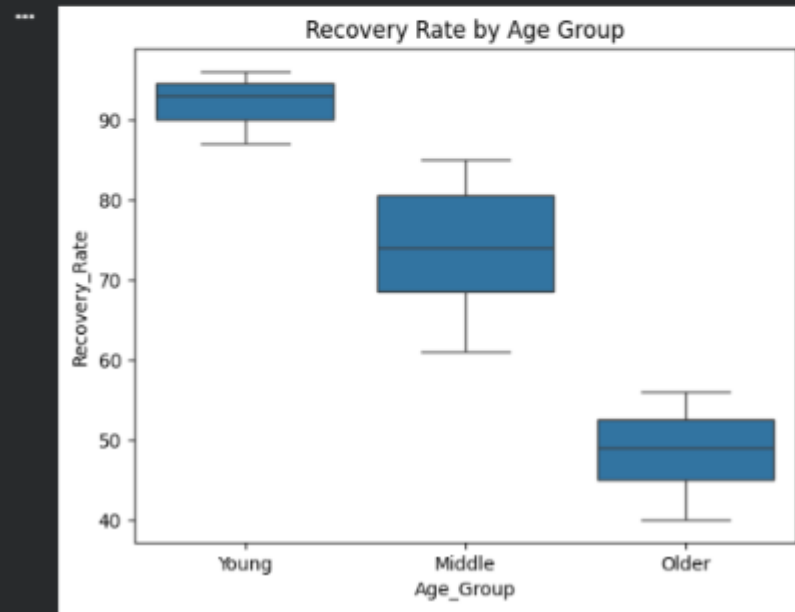
```

df["Age_Group"] = pd.cut(
    df["Age"],
    bins=[0, 40, 60, 100],
    labels=["Young", "Middle", "Older"]
)

sns.boxplot(
    data=df,
    x="Age_Group",
    y="Recovery_Rate"
)

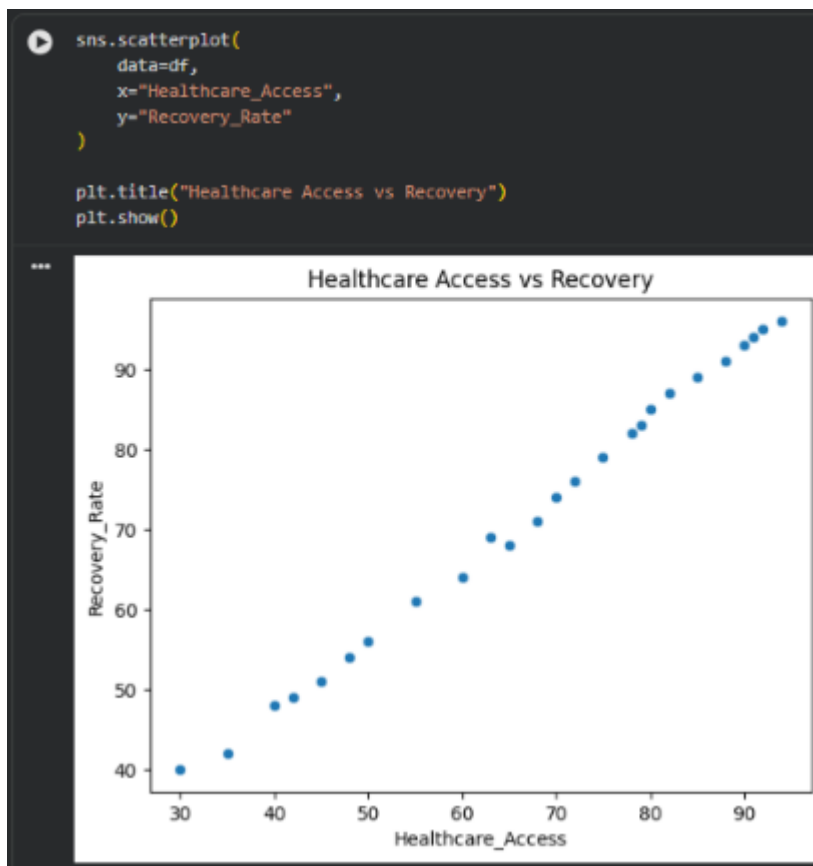
plt.title("Recovery Rate by Age Group")
plt.show()

```



This graph can reveal that age itself may influence recovery.

Graph 3 — Healthcare Access vs Recovery



This helps identify another possible explanation for recovery differences.

Evaluation Table

Parameter	Result/Interpretation
Accuracy	Measures correct classifications
Precision	Measures correctness of positive predictions
Recall	Measures detected high-recovery patients
F1 Score	Balance between precision and recall
Exercise correlation	Positive association
Age effect	Possible confounder
Healthcare access effect	Possible confounder
Causal relationship	Not established

Example format for model results:

Metric	Value
--------	-------

Accuracy	0.89
----------	------

Precision	0.87
-----------	------

Recall	0.90
--------	------

F1 Score	0.88
----------	------

Systematic Validation Framework

Before implementing the national intervention:

Step 1 — Data Quality

Check:

- Missing values
- Duplicate records
- Incorrect measurements
- Outliers
- Inconsistent patient records

Step 2 — Exploratory Analysis

Perform:

- Scatter plots
- Correlation analysis
- Box plots
- Group comparisons
- Distribution analysis

Step 3 — Confounding Analysis

Check whether the relationship changes after considering:

- Age
- Income
- Healthcare access

- Treatment quality
- Lifestyle
- Disease severity

Step 4 — Statistical Testing

Test whether the observed relationship is statistically significant.

Step 5 — Machine Learning

Compare models using:

- Accuracy
- Precision
- Recall
- F1-score
- Cross-validation

Step 6 — Causal Validation

Use appropriate approaches such as:

- Controlled experiments
- Randomized trials where ethical and feasible
- Longitudinal studies
- Causal inference methods

Step 7 — Policy Pilot

Instead of immediately implementing a nationwide program:

Dataset → Analysis → Validation → Small pilot → Evaluate outcome → Large-scale implementation

Final Comparison of the 4 Questions

Question Dataset		Model	Main Focus	Main Metrics
Q1	University Placement	Linear Regression	Correlation vs causation	R^2 , RMSE

Question Dataset		Model	Main Focus	Main Metrics
Q5	University Placement	Random Forest	Investment decision	Accuracy, Precision, Recall, F1
Q6	Healthcare	Random Forest Regression	Exercise vs recovery	R^2 , RMSE
Q10	Healthcare	Logistic Regression	Confounding & policymaking	Accuracy, Precision, Recall, F1

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